

BUSFIN 8200: Asset Pricing Foundations

Problem Set 1

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1 Question 1

1.1 Derivation of Equations 1.1–1.3

Task. (1) Starting from $R_{e,t+1} = (P_{t+1} + D_{t+1})/P_t$, derive the (approximate) Equations 1.1, 1.2, and 1.3.

Throughout, a lowercase letter denotes the log of the corresponding uppercase variable, e.g. $r_{e,t+1} = \log(R_{e,t+1})$, $p_t = \log(P_t)$ and $d_t = \log(D_t)$. The log dividend–price ratio and the log dividend growth are $dp_t := d_t - p_t$ and $\Delta d_{t+1} := d_{t+1} - d_t$.

Equation 1.1. Start from the definition of the gross equity return,

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}. \quad (1)$$

Taking logs of (1),

$$\begin{aligned} \log(R_{e,t+1}) &= \log(P_{t+1}) - \log(P_t) + \log\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) \\ \iff r_{e,t+1} &= p_{t+1} - p_t + \log\left(1 + e^{dp_{t+1}}\right). \end{aligned} \quad (2)$$

Rearranging (2) so that the linear terms are on the left-hand side and the log term is on the right-hand side,

$$r_{e,t+1} - p_{t+1} + p_t = \log\left(1 + e^{dp_{t+1}}\right). \quad (3)$$

The function $\log(1 + e^x)$ can be approximated by a first-order Taylor expansion around a point $\overline{dp}$:

$$\log(1 + e^x) \approx \log\left(1 + e^{\overline{dp}}\right) + \frac{e^{\overline{dp}}}{1 + e^{\overline{dp}}} (x - \overline{dp}), \quad (4)$$

which is accurate for small enough $x - \overline{dp}$. The expansion point is taken to be $\overline{dp} = \mathbb{E}[dp_t]$. Substituting (4) into the right-hand side of (3), with $x = dp_{t+1}$, and rearranging the terms, we obtain

$$p_t = \kappa_0 + (1 - \kappa) d_{t+1} - r_{e,t+1} + \kappa p_{t+1}, \quad (5)$$

where, as in the problem statement,

$$\kappa = \frac{1}{1 + e^{\overline{dp}}} \quad \text{and} \quad \kappa_0 = -\log(\kappa) - (1 - \kappa) \log\left(\frac{1}{\kappa} - 1\right). \quad (6)$$

Subtracting d_t from both sides of (5), multiplying both sides by -1 , and using the definitions of dp_t and Δd_{t+1} , we obtain the one-period identity

$$dp_t = -\kappa_0 + \kappa dp_{t+1} + r_{e,t+1} - \Delta d_{t+1}. \quad (7)$$

Equation (7) holds at every date. Iterating (7) forward H times, i.e. substituting the future dp terms up to horizon H , we get

$$dp_t = -\kappa_0 \sum_{h=0}^{H-1} \kappa^h + \kappa^H dp_{t+H} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h}, \quad (8)$$

where the constant term is a finite geometric sum,

$$\sum_{h=0}^{H-1} \kappa^h = \frac{1 - \kappa^H}{1 - \kappa}, \quad (9)$$

with $0 < \kappa < 1$ because $\kappa = 1/(1 + e^{\bar{dp}})$ and $e^{\bar{dp}} > 0$. Equation (8) with the constant written as in (9) is Equation 1.1 of the problem statement, since $-\kappa_0(1 - \kappa^H)/(1 - \kappa) = \kappa_0(\kappa^H - 1)/(1 - \kappa)$.

Equation 1.2. Taking the expectation conditional on the information available at time t , denoted $\mathbb{E}_t[\cdot]$, on both sides of (8), using the linearity of the expectation and $\mathbb{E}_t[dp_t] = dp_t$, and using (9), we get

$$dp_t = -\kappa_0 \frac{1 - \kappa^H}{1 - \kappa} + \kappa^H \mathbb{E}_t[dp_{t+H}] + \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}], \quad (10)$$

which is Equation 1.2 of the problem statement.

Equation 1.3. Letting $H \rightarrow \infty$ in (10) requires the following assumptions.

(i) The no-bubble condition

$$\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t[dp_{t+H}] = 0. \quad (11)$$

(ii) r_e and Δd are covariance stationary with finite second moments, which implies that the two infinite sums in (10) converge absolutely almost surely.¹

Under (i)–(ii), the limit of (10) is

$$dp_t = -\frac{\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}], \quad (12)$$

which is Equation 1.3 of the problem statement.

¹For $x \in \{r_e, \Delta d\}$: $\mathbb{E}|x_{t+h}| \leq (\mathbb{E}x^2)^{1/2} =: C$ by covariance stationarity, and $|\mathbb{E}_t[x_{t+h}]| \leq \mathbb{E}_t|x_{t+h}|$ by conditional Jensen. Hence $\mathbb{E}[\sum_{h \geq 1} \kappa^{h-1} \mathbb{E}_t|x_{t+h}|] = \sum_{h \geq 1} \kappa^{h-1} \mathbb{E}_t|x_{t+h}| \leq C/(1 - \kappa) < \infty$ (monotone convergence, iterated expectations), so the sum is a.s. finite and $\sum_{h \geq 1} \kappa^{h-1} \mathbb{E}_t[x_{t+h}]$ converges absolutely a.s.

Task. (2) Explain and justify any assumptions you make.

Three assumptions enter the derivation.

Log-linearization. The only approximation is (4), whose error is of second order in $dp_{t+1} - \overline{dp}$. The approximation is accurate when dp is close to its mean and less accurate far from it.

No bubble (i). Condition (11) rules out rational bubbles, i.e. paths on which P/D grows explosively and becomes detached from expected dividends and returns.

Stationarity (ii). Covariance stationarity of r_e and Δd ensures that the infinite sums converge (footnote 1). The assumption is mild for these variables, which have no trend and are much less persistent than dp .

1.2 Variance decomposition by horizon (data)

Task. (1) Applying Equation 1.4 to the EQ Dataset, reproduce the figure provided on slide 1.8 of the Module 1 lecture notes. This figure has H in the x-axis and $(b_{r_e}^{(H)}, b_{\Delta d}^{(H)}, b_{dp}^{(H)})$ in the y-axis.

Equation 1.4 of the problem statement splits the variance of dp_t into three parts,

$$1 = \underbrace{\frac{\text{Cov}\left[dp_t, \sum_{h=1}^H \kappa^{h-1} r_{e,t+h}\right]}{\text{Var}[dp]}}_{b_{r_e}^{(H)}} + \underbrace{\frac{\text{Cov}\left[dp_t, -\sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h}\right]}{\text{Var}[dp]}}_{b_{\Delta d}^{(H)}} + \underbrace{\frac{\text{Cov}\left[dp_t, \kappa^H dp_{t+H}\right]}{\text{Var}[dp]}}_{b_{dp}^{(H)}}. \quad (13)$$

The coefficients are estimated on the EQ Dataset as follows.

- *Discount coefficient.* $\kappa = 1/(1 + e^{\overline{dp}})$, where $\overline{dp}$ is the full-sample mean of dp_t (1927:12–2021:12). This gives $\kappa = 0.9642$, which is used for every H .
- *Timing.* The data are monthly observations of annual variables, so $t + h$ is $12h$ months after t .
- *Estimation.* For each $H = 1, \dots, 15$, each of the three terms is regressed by OLS on a constant and dp_t . With a constant, the OLS slope equals the sample covariance divided by the sample variance of dp_t , i.e. the corresponding coefficient in (13).
- *Sample.* Each regression uses every month t for which $t + 12H$ is in the data, so the three terms share one sample for each H (949 to 1,117 observations).

Figure 1 plots the three coefficients against H , and Table 1 reports them with their sum (code: `src/q1b.py`).

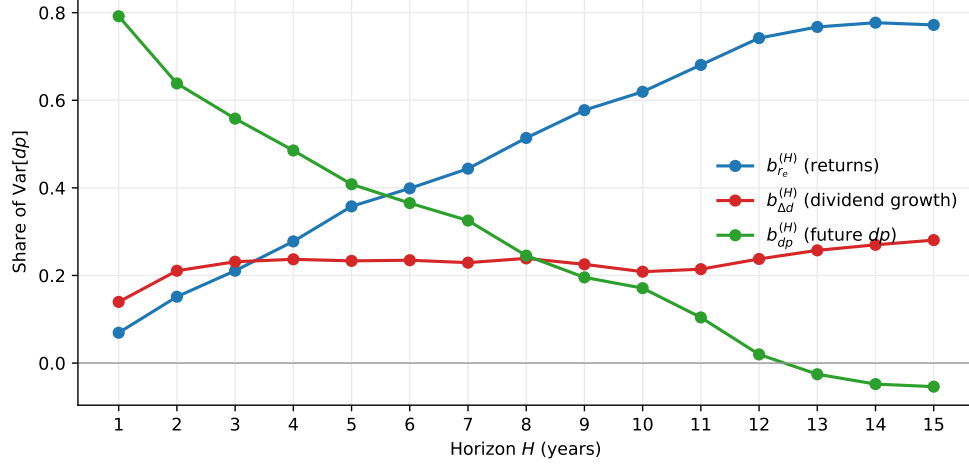


Figure 1: Coefficients $b_{re}^{(H)}$, $b_{\Delta d}^{(H)}$ and $b_{dp}^{(H)}$ of the decomposition (13) against the horizon H (in years), estimated on the EQ Dataset with $\kappa = 0.9642$. Monthly observations, 1927:12–2021:12.

H	$b_{re}^{(H)}$	$b_{\Delta d}^{(H)}$	$b_{dp}^{(H)}$	Sum	N
1	0.069	0.140	0.792	1.001	1,117
2	0.152	0.211	0.638	1.001	1,105
3	0.211	0.231	0.558	1.000	1,093
4	0.278	0.237	0.485	1.000	1,081
5	0.358	0.233	0.408	0.999	1,069
6	0.399	0.235	0.365	0.999	1,057
7	0.444	0.229	0.325	0.998	1,045
8	0.514	0.239	0.245	0.998	1,033
9	0.577	0.225	0.196	0.998	1,021
10	0.619	0.209	0.171	0.999	1,009
11	0.681	0.214	0.104	0.999	997
12	0.742	0.238	0.020	0.999	985
13	0.767	0.257	-0.026	0.999	973
14	0.777	0.270	-0.048	0.999	961
15	0.772	0.281	-0.054	0.999	949

Table 1: Coefficients of the decomposition (13) by horizon H , their sum, and the number of monthly observations N (predictor months from 1927:12 to $12H$ months before 2021:12).

Task. (2) Explain what the inference from this figure is.

Observation. As H increases, the share of future dp falls, while the share of returns rises (to about 0.77 at $H = 15$, against about 0.28 for dividend growth).

Inference. This implies that most of the variation in dp reflects changes in expected future returns rather than in expected dividend growth.

1.3 Variance decomposition by horizon (VAR-implied)

Task. Now, use Equation 1.4 and the EQ Dataset to reproduce a figure analogous to the one in Question 1b) but with the terms from Equation 1.4 implied by the VAR (estimated by OLS) $z_{t+1} = \Gamma_0 + \Gamma z_t + \tilde{z}_{t+1}$ where $\tilde{z}_{t+1} \sim IID(0, \Sigma)$ (1.5) where $z_t = [\Delta d_t \ r_{e,t} \ dp_t]$.

With $\mu = (I - \Gamma)^{-1}\Gamma_0$, iterating (1.5) forward h times gives

$$z_{t+h} - \mu = \Gamma^h(z_t - \mu) + \sum_{j=1}^h \Gamma^{h-j} \tilde{z}_{t+j}, \quad \text{so} \quad \mathbb{E}_t[z_{t+h}] - \mu = \Gamma^h(z_t - \mu). \quad (14)$$

Since future shocks are uncorrelated with z_t ,

$$\text{Cov}[z_{t+h}, dp_t] = \Gamma^h \Sigma_z e_{dp}, \quad \text{where} \quad \Sigma_z = \Gamma \Sigma_z \Gamma' + \Sigma, \quad (15)$$

and $e_{\Delta d}$, e_r , e_{dp} select the elements of z . Substituting into Equation 1.4 gives

$$b_{r_e}^{(H)} = \frac{\sum_{h=1}^H \kappa^{h-1} e_r' \Gamma^h \Sigma_z e_{dp}}{\text{Var}[dp]}, \quad b_{\Delta d}^{(H)} = -\frac{\sum_{h=1}^H \kappa^{h-1} e_{\Delta d}' \Gamma^h \Sigma_z e_{dp}}{\text{Var}[dp]}, \quad b_{dp}^{(H)} = \frac{\kappa^H e_{dp}' \Gamma^H \Sigma_z e_{dp}}{\text{Var}[dp]}, \quad (16)$$

with the VAR-implied variance $\text{Var}[dp] = e_{dp}' \Sigma_z e_{dp}$.

The VAR is estimated on the EQ Dataset as follows.

- *Timing.* One step of the VAR is 12 months, as one year h in Question 1b: z at month $t + 12$ is regressed on z at month t .
- *Estimation.* OLS equation by equation, with a constant, on every month t for which $t + 12$ is in the data (1,117 observations, 1927:12–2020:12). Σ is the covariance matrix of the residuals, and Σ_z solves (15).
- *Stationarity.* The largest eigenvalue of Γ has modulus 0.869, so the VAR is stationary and (15) has a unique solution.
- *Terms.* The same $\kappa = 0.9642$ and $H = 1, \dots, 15$ as in Question 1b, in (16).

Figure 2 plots the three coefficients against H on the same vertical scale as Figure 1, and Table 2 reports them with their sum (code: `src/q1c.py`).

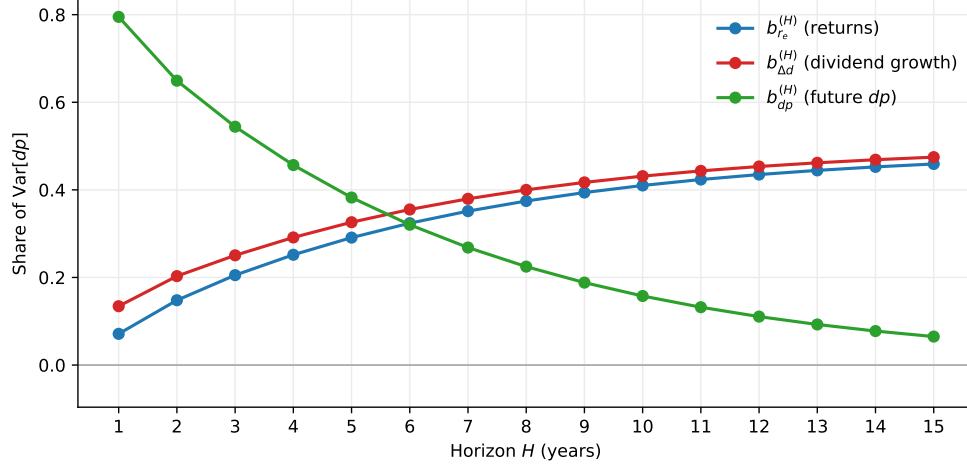


Figure 2: VAR-implied coefficients $b_{re}^{(H)}$, $b_{\Delta d}^{(H)}$ and $b_{dp}^{(H)}$ of (16) against the horizon H (in years), with $\kappa = 0.9642$. VAR (1.5) with a 12-month step, estimated by OLS on monthly observations, 1927:12–2021:12.

H	$b_{re}^{(H)}$	$b_{\Delta d}^{(H)}$	$b_{dp}^{(H)}$	Sum	N
1	0.071	0.134	0.795	1.001	1,117
2	0.148	0.203	0.649	1.000	1,117
3	0.205	0.251	0.544	1.000	1,117
4	0.252	0.291	0.457	1.000	1,117
5	0.291	0.326	0.382	1.000	1,117
6	0.324	0.355	0.320	0.999	1,117
7	0.351	0.380	0.268	0.999	1,117
8	0.375	0.400	0.225	0.999	1,117
9	0.394	0.417	0.188	0.999	1,117
10	0.410	0.431	0.158	0.999	1,117
11	0.424	0.443	0.132	0.999	1,117
12	0.435	0.453	0.111	0.999	1,117
13	0.444	0.462	0.093	0.999	1,117
14	0.452	0.469	0.078	0.999	1,117
15	0.459	0.475	0.065	0.999	1,117

Table 2: VAR-implied coefficients of (16) by horizon H and their sum. N is the number of monthly observations used to estimate the VAR, the same for every H .

1.4 Infinite-horizon decomposition

Task. (1) Using the VAR in Equation 1.5, calculate the values for $b_{r_e}^{(\infty)}$ and $b_{\Delta d}^{(\infty)}$.

Taking $H \rightarrow \infty$, Equation 1.4 becomes Equation 1.6 of the problem statement,

$$1 = \underbrace{\frac{\text{Cov}[dp_t, \sum_{h=1}^{\infty} \kappa^{h-1} r_{e,t+h}]}{\text{Var}[dp]}}_{b_{r_e}^{(\infty)}} + \underbrace{\frac{\text{Cov}[dp_t, -\sum_{h=1}^{\infty} \kappa^{h-1} \Delta d_{t+h}]}{\text{Var}[dp]}}_{b_{\Delta d}^{(\infty)}}. \quad (17)$$

With the VAR of Question 1c, the covariances follow from (15) and

$$\sum_{h=1}^{\infty} \kappa^{h-1} \Gamma^h = \Gamma(I - \kappa \Gamma)^{-1}, \quad (18)$$

which holds when all eigenvalues of $\kappa \Gamma$ lie inside the unit circle. This gives

$$b_{r_e}^{(\infty)} = \frac{e'_r \Gamma(I - \kappa \Gamma)^{-1} \Sigma_z e_{dp}}{\text{Var}[dp]}, \quad b_{\Delta d}^{(\infty)} = -\frac{e'_{\Delta d} \Gamma(I - \kappa \Gamma)^{-1} \Sigma_z e_{dp}}{\text{Var}[dp]}, \quad (19)$$

with Γ , Σ_z , $\kappa = 0.9642$ and the VAR-implied $\text{Var}[dp] = e'_{dp} \Sigma_z e_{dp}$ of Question 1c. The largest eigenvalue of $\kappa \Gamma$ has modulus 0.838, so (18) applies. Table 3 reports the two coefficients (code: `src/q1d.py`).

Coefficient	Value
$b_{r_e}^{(\infty)}$	0.494
$b_{\Delta d}^{(\infty)}$	0.505
Sum	0.999

Table 3: VAR-implied infinite-horizon coefficients of (17), computed from (19) with the VAR of Question 1c (12-month step, estimated by OLS on monthly observations, 1927:12–2021:12) and $\kappa = 0.9642$.

Task. (2) Interpret the results.

Result. In the VAR, as $H \rightarrow \infty$, expected returns and expected dividend growth each account for about half of the variance of dp . Together, they account for all of it.

Comparison. The two methods already differ at horizons within the sample (e.g. $H = 5$ to 15), so the difference comes from the VAR(1) restriction: the VAR computes all covariances from one-year dynamics instead of estimating them directly.

1.5 Derivation of Equations 1.7–1.9 (Gao and Martin, 2021)

Task. (1) Starting from $R_{e,t+1} = (P_{t+1} + D_{t+1})/P_t$, derive the (approximate) Equations 1.7, 1.8, and 1.9.

As in the problem statement, $dy_t = \log(1 + D_t/P_t)$ and $\kappa = e^{-\overline{dy}}$.

Equation 1.7. Write the gross equity return as

$$R_{e,t+1} = \frac{D_{t+1}}{D_t} \frac{D_t}{P_t} \left(1 + \frac{P_{t+1}}{D_{t+1}} \right). \quad (20)$$

Since $1 + D_t/P_t = e^{dy_t}$, we have

$$\frac{D_t}{P_t} = e^{dy_t} - 1 = e^{dy_t} (1 - e^{-dy_t}) \quad \text{and} \quad 1 + \frac{P_{t+1}}{D_{t+1}} = \frac{1 + D_{t+1}/P_{t+1}}{D_{t+1}/P_{t+1}} = \frac{1}{1 - e^{-dy_{t+1}}}. \quad (21)$$

Substituting the two relations above into (20) and taking logs, we get

$$r_{e,t+1} - \Delta d_{t+1} = dy_t + \log(1 - e^{-dy_t}) - \log(1 - e^{-dy_{t+1}}). \quad (22)$$

A first-order Taylor expansion of $\log(1 - e^{-x})$ around $\bar{dy}$ gives

$$\log(1 - e^{-x}) \approx \log(1 - \kappa) + \frac{\kappa}{1 - \kappa} (x - \bar{dy}). \quad (23)$$

Substituting (23) into both log terms of (22), the constants cancel, and rearranging gives the one-period identity, with no constant term,

$$dy_t = (1 - \kappa) (r_{e,t+1} - \Delta d_{t+1}) + \kappa dy_{t+1}. \quad (24)$$

Iterating (24) forward H times, we obtain Equation 1.7 of the problem statement,

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right) + \kappa^H dy_{t+H}. \quad (25)$$

Equation 1.8. Taking $\mathbb{E}_t[\cdot]$ on both sides of (25), using the linearity of the expectation and $\mathbb{E}_t[dy_t] = dy_t$, as in Section 1.1, we get Equation 1.8 of the problem statement,

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] \right) + \kappa^H \mathbb{E}_t[dy_{t+H}]. \quad (26)$$

Equation 1.9. Let $H \rightarrow \infty$ in (26) under the following assumptions.

(i) The no-bubble condition

$$\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t[dy_{t+H}] = 0. \quad (27)$$

(ii) r_e and Δd are covariance stationary (see footnote 1).

Note that $0 < \kappa < 1$ because $dy > 0$. The limit of (26) is Equation 1.9 of the problem statement,

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] \right). \quad (28)$$

Task. (2) Explain and justify any assumptions you make.

Three assumptions enter the derivation.

Log-linearization. The only approximation is (23). Unlike $\log(D/P)$ in Section 1.1, $dy = \log(1 + D/P)$ is close to D/P itself when D/P is small, so it moves much less when D/P is far from its mean (e.g. in the late 1990s). This is why Gao and Martin (2021) argue that the approximation is more accurate there.

No bubble (i). Condition (27) is the same as the no-bubble condition of Section 1.1.

Stationarity (ii). This is the same as assumption (ii) of Section 1.1 (footnote 1).

2 Question 2

2.1 Predictive regressions by horizon

Task. (1) Using the EQ Dataset, estimate the regression (by OLS) $\frac{1}{H} \cdot \sum_{h=1}^H xR_{e,t+h} = a^{(H)} + b^{(H)} \cdot D_t/P_t + \varepsilon_t^{(H)}$ (2.1) for $H = 1, 2, 3, \dots, 14, 15$ years (with $xR_{e,t} = e^{r_{e,t}} - e^{r_{f,t}}$ and $D_t/P_t = e^{dp_t}$). Plot a graph with H in the x-axis and the R_{adj}^2 of these regressions in the y-axis.

Figure 3 plots the adjusted R^2 of the regression

$$\frac{1}{H} \sum_{h=1}^H xR_{e,t+h} = a^{(H)} + b^{(H)} \frac{D_t}{P_t} + \varepsilon_t^{(H)}, \quad H = 1, \dots, 15, \quad (29)$$

estimated by OLS on the EQ Dataset, where $xR_{e,t} = e^{r_{e,t}} - e^{r_{f,t}}$ and $D_t/P_t = e^{dp_t}$. The data are monthly observations of annual variables, so $t + h$ is $12h$ months after t , and each horizon uses every month t for which H years of subsequent data are available (code: `src/q2a.py`).

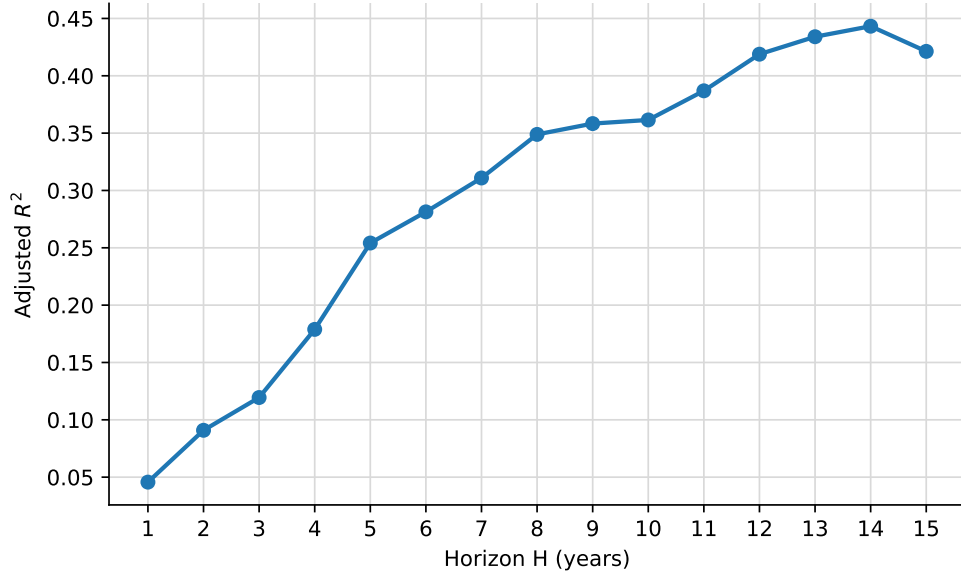


Figure 3: Adjusted R^2 of regression (29) against the horizon H (in years). Monthly observations, 1927:12–2021:12.

Task. (2) Describe the results you observe.

Observation. The adjusted R^2 increases from $H = 1$ to $H = 15$.

Inference. D/P predicts long-horizon returns much better than one-year returns. Because D/P is persistent, expected returns move slowly, so their effect can accumulate over long horizons.

2.2 Standard errors for the one-year predictive regression

Task. Estimate the regression (by OLS) $xR_{e,t+1} = a + b \cdot D_t/P_t + \varepsilon_t$ (2.2) but this time also obtain standard errors for $\hat{b}$ using five different methods:

- (i) The baseline OLS standard errors
- (ii) The White (1980) standard errors
- (iii) The Newey and West (1987) standard errors with 11 lags
- (iv) The Hansen and Hodrick (1980) standard errors with 11 lags
- (v) The Newey and West (1987, 1994) standard errors

[...] Report $\hat{b}$ as well as its t -statistic based on each of the five standard error estimation methods above.

Table 4 reports the OLS estimates of

$$xR_{e,t+1} = a + b \frac{D_t}{P_t} + \varepsilon_t, \quad (30)$$

together with the standard error and t -statistic of $\hat{b}$ under five standard-error methods. As in Question 2a, $xR_{e,t} = e^{r_{e,t}} - e^{r_{f,t}}$ and $D_t/P_t = e^{dp_t}$; $t + 1$ is 12 months after t , and the sample uses every month t for which the subsequent annual return is available (code: `src/q2b.py`).

Standard-error method	SE($\hat{b}$)	$t(\hat{b})$	Lags
(i) OLS	0.3802	7.38	–
(ii) White (1980)	0.6587	4.26	–
(iii) Newey-West (1987), 11 lags	1.2979	2.16	11
(iv) Hansen-Hodrick (1980), 11 lags	1.4521	1.93	11
(v) Newey-West (1987, 1994), automatic lags	1.2311	2.28	24
$\hat{a} = -0.0314$, $\hat{b} = 2.8038$, $T = 1117$ monthly observations			

Table 4: OLS estimates of regression (30) with the standard error and t -statistic of $\hat{b}$ under five standard-error methods. Predictor dates 1927:12–2020:12. “Lags” is the number of autocorrelation lags; for method (v) it is selected by the Newey–West (1994) automatic bandwidth rule.

2.3 Amihud and Hurvich (2004) bias-corrected estimate

Task. (1) Reestimate the regression in Equation 2.2 but this time using the method in Amihud and Hurvich (2004). Report the b estimate obtained in this question and contrast it to the b estimated obtained in Question (2b).

Table 5 reports the Amihud and Hurvich (2004) estimate $\hat{b}^c$ of b in regression (30), on the same sample as Table 4. First, D_{t+1}/P_{t+1} is regressed on D_t/P_t by OLS, giving $\hat{\theta}$ and $\hat{\rho}$, and the

slope is bias-corrected as $\hat{\rho}^c = \hat{\rho} + (1 + 3\hat{\rho})/n + 3(1 + 3\hat{\rho})/n^2$ with $n = 94$ years. Second, with $\hat{u}_{t+1}^c = D_{t+1}/P_{t+1} - (\hat{\theta} + \hat{\rho}^c D_t/P_t)$,

$$xR_{e,t+1} = a + b \frac{D_t}{P_t} + b_u \hat{u}_{t+1}^c + \varepsilon_t \quad (31)$$

is estimated by OLS, and $\hat{b}^c$ is the coefficient on D_t/P_t . Its standard error follows Amihud and Hurvich (2004, Eq. 10), $\widehat{SE}(\hat{b}^c) = \sqrt{(\hat{b}_u)^2 (1 + 3/n + 9/n^2)^2 \widehat{\text{Var}}(\hat{\rho}) + \widehat{SE}(\hat{b}^c)^2}$ with $n = 94$ (code: `src/q2c.py`).

Standard error of $\hat{b}^c$	SE($\hat{b}^c$)	$t(\hat{b}^c)$
(i-a) Eq. (10), OLS ingredients	0.3876	6.03
(i-b) Eq. (10), Newey-West (1987) 11 lags ingredients	1.5170	1.54
$\hat{b}^c = 2.3362, \quad \hat{a} = -0.0314, \quad \hat{b}_u = -13.4837$		
$\hat{\rho} = 0.7197, \quad \hat{\rho}^c = 0.7544, \quad n = 94 \text{ years}, \quad T = 1117 \text{ monthly observations}$		
OLS $\hat{b}$ (Question 2b) = 2.8038		

Table 5: Amihud and Hurvich (2004) estimate of b in regression (30) via the augmented regression (31). In row (i-a), $\widehat{\text{Var}}(\hat{\rho})$ and $\widehat{SE}(\hat{b}^c)$ are plain OLS estimates; in row (i-b), both are Newey–West (1987) estimates with 11 lags. Predictor dates 1927:12–2020:12.

Task. (2) Explain why it is natural for the two estimates to differ.

OLS is biased here because of the Stambaugh bias. In particular, D/P is very persistent, so OLS underestimates its persistence in small samples. Because D/P shocks and return shocks are negatively correlated ($\hat{b}_u < 0$), this error pushes $\hat{b}$ up. The Amihud–Hurvich method corrects for this bias, so $\hat{b}^c$ is smaller.

2.4 Out-of-sample estimation (expanding window)

Task. (1) Estimate the regression in Equation 2.2 again, but this time using out-of-sample (OS) estimation with an expanding window (i.e., using time-varying a_t and b_t that are estimated with historical data). Initiate your OS estimation in December/1940 (i.e., D_t/P_t in December/1939 and $xR_{e,t+1}$ in December/1940). Label the historical (expanding window) mean as $\overline{xR}_{e,t}$, the fitted values from Question (1b) as $\hat{E}_t^{IS}[xR_e]$, and the OS forecasts from this question as $\hat{E}_t^{OS}[xR_e]$. Plot the time-series of $\overline{xR}_{e,t}$, $\hat{E}_t^{IS}[xR_e]$, and $\hat{E}_t^{OS}[xR_e]$ from December/1940 to the end of the sample.

At each forecast origin t , only the information available at t is used. Regression (30) is estimated by OLS on the pairs $(D_s/P_s, xR_{e,s+1})$ whose return is realized by t , i.e. with s at least 12 months before t , starting with the predictor of 1927:12. This gives the time-varying estimates $\hat{a}_t$ and $\hat{b}_t$ and the out-of-sample forecast

$$\hat{E}_t^{OS}[xR_{e,t+1}] = \hat{a}_t + \hat{b}_t \frac{D_t}{P_t}. \quad (32)$$

The benchmark forecast is the historical mean of all annual excess returns realized by t ,

$$\overline{xR}_{e,t} = \frac{1}{n_t} \sum_{m \leq t} xR_{e,m}, \quad (33)$$

where m runs over the n_t months from 1927:12 to t ; the return dated 1927:12 is included in the mean although it never enters the regression, because no D/P is observed 12 months before it. The in-sample fitted value is $\hat{E}_t^{IS}[xR_{e,t+1}] = \hat{a} + \hat{b} D_t/P_t$, with $\hat{a}$ and $\hat{b}$ from the full sample of Question 2b. The forecast origins are $t = 1939:12, \dots, 2020:12$, so the forecast targets $t + 1$ are the 973 months 1940:12–2021:12 (code: `src/q2d.py`).

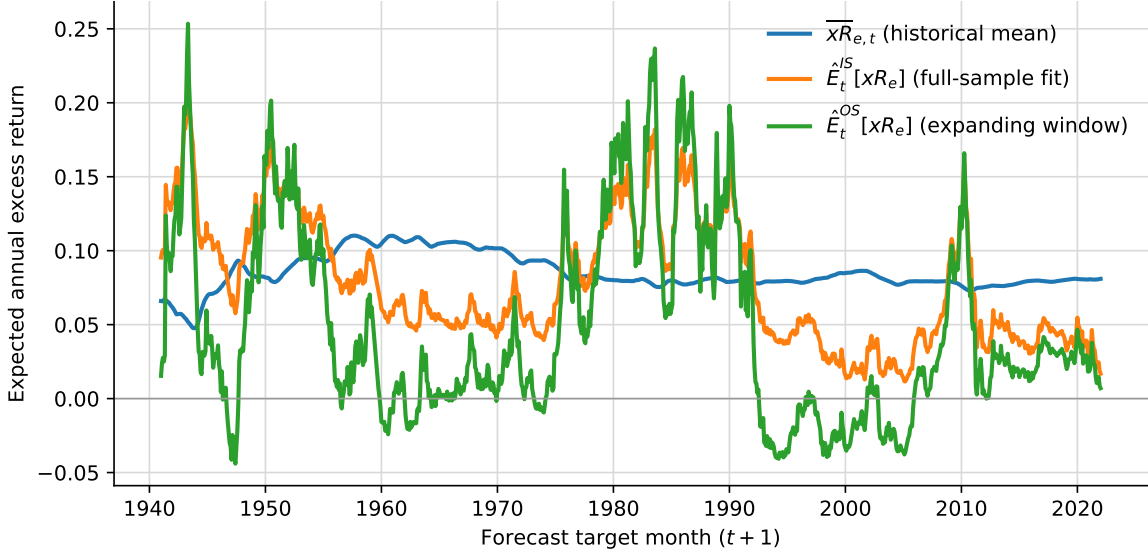


Figure 4: Expanding-window historical mean $\overline{xR_{e,t}}$, in-sample fitted values $\hat{E}_t^{IS}[xR_e]$ from the full-sample regression of Question 2b, and out-of-sample forecasts $\hat{E}_t^{OS}[xR_e]$ from regression (30) estimated with data available at t , plotted against the forecast-target month $t + 1$, 1940:12–2021:12.

Task. (2) Then, calculate and report the R_{OS}^2 value using the forecasting errors from December/1940 to the end of the sample (note that we calculate R_{OS}^2 instead of $\overline{R}_{OS}^2$ because a degree of freedom adjustment is not needed in an OS analysis).

The out-of-sample R^2 compares the squared forecast errors of (32) with those of the historical mean (33),

$$R_{OS}^2 = 1 - \frac{\sum_{t \in \mathcal{T}} (xR_{e,t+1} - \hat{E}_t^{OS}[xR_{e,t+1}])^2}{\sum_{t \in \mathcal{T}} (xR_{e,t+1} - \overline{xR_{e,t}})^2}, \quad (34)$$

where $\mathcal{T}$ is the set of forecast origins. Over all origins, i.e. over the forecast-target months 1940:12–2021:12, $R_{OS}^2 = -0.0135$.

Task. (3) Now, to understand how R_{OS}^2 varies over time, calculate R_{OS}^2 using a 50-year rolling window for the forecasting errors (without modifying the a_t and b_t calculations) and provide a time-series plot of these R_{OS}^2 values (note that the time-series will start in December/1990 and end at the end of your sample).

For the rolling version, $\hat{a}_t$, $\hat{b}_t$ and $\overline{xR_{e,t}}$ are unchanged and only the two sums in (34) are restricted to a window: $R_{OS}^2(T)$ uses the 600 consecutive forecast-target months ending at T , for $T = 1990:12, \dots, 2021:12$ (the first window is 1941:01–1990:12).

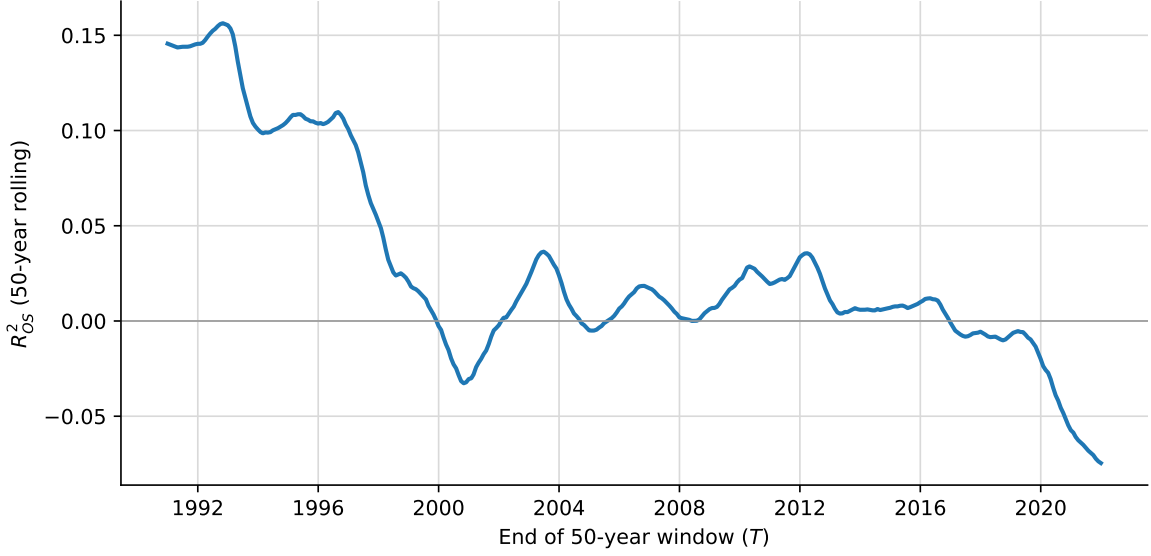


Figure 5: R^2_{OS} computed over a rolling window of 600 forecast-target months ending at T , for T from 1990:12 to 2021:12, with $\hat{a}_t$, $\hat{b}_t$ and $\overline{xR}_{e,t}$ unchanged.

2.5 Out-of-sample estimation with steady-state restrictions

Task. Repeat Question (2d). However, this time estimate the OS coefficients using $\hat{a}_t = G_t - 1$ and $\hat{b}_t = G_t$ when obtaining $\hat{E}_t^{OS}[xR_e]$, where G_t is the historical (expanding window) average of $e^{\Delta d_t}$ over the same sample used to calculate $\overline{xR}_{e,t}$.

The procedure is that of Question 2d except for the coefficients, which are restricted to $\hat{a}_t = G_t - 1$ and $\hat{b}_t = G_t$, where

$$G_t = \frac{1}{n_t} \sum_{m \leq t} e^{\Delta d_m} \quad (35)$$

is the mean of the gross dividend growth over the n_t months m from 1927:12 to t , the same months whose returns enter $\overline{xR}_{e,t}$ in (33). The out-of-sample forecast is

$$\hat{E}_t^{OS}[xR_{e,t+1}] = (G_t - 1) + G_t \frac{D_t}{P_t}, \quad (36)$$

and R^2_{OS} is computed from (34) with this forecast, with the same historical mean, forecast origins, forecast targets and 600-month rolling windows as in Question 2d (code: `src/q2e.py`).

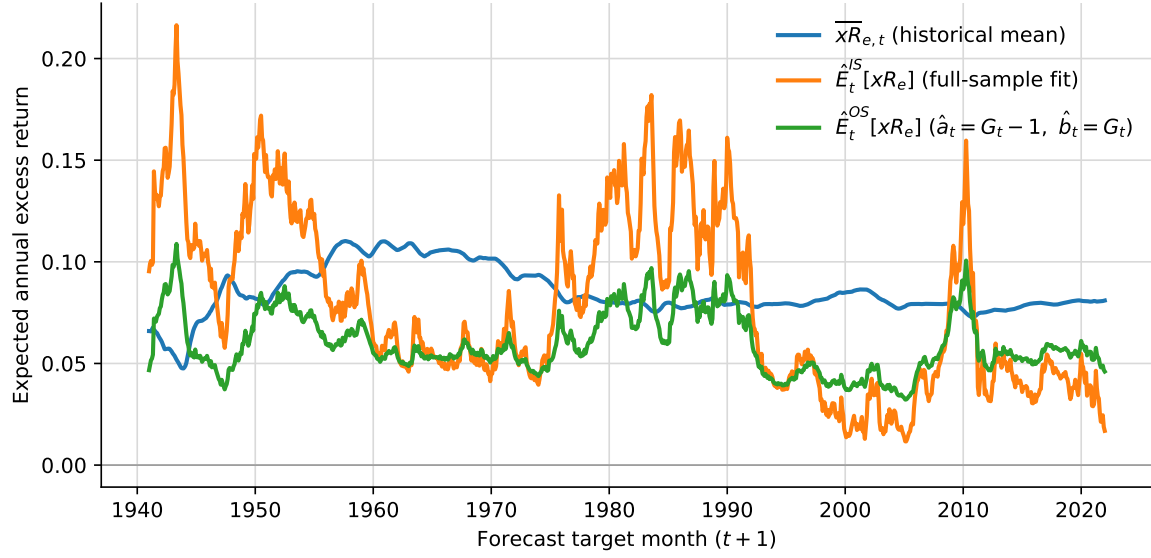


Figure 6: Expanding-window historical mean $\overline{xR_{e,t}}$, in-sample fitted values $\hat{E}_t^{IS}[xR_e]$ from the full-sample regression of Question 2b, and out-of-sample forecasts $\hat{E}_t^{OS}[xR_e] = (G_t - 1) + G_t D_t/P_t$, where G_t is the expanding mean of $e^{\Delta d}$ over 1927:12 to t , plotted against the forecast-target month $t + 1$, 1940:12–2021:12.

The out-of-sample R^2 over the forecast-target months 1940:12–2021:12 is $R_{OS}^2 = 0.0294$.

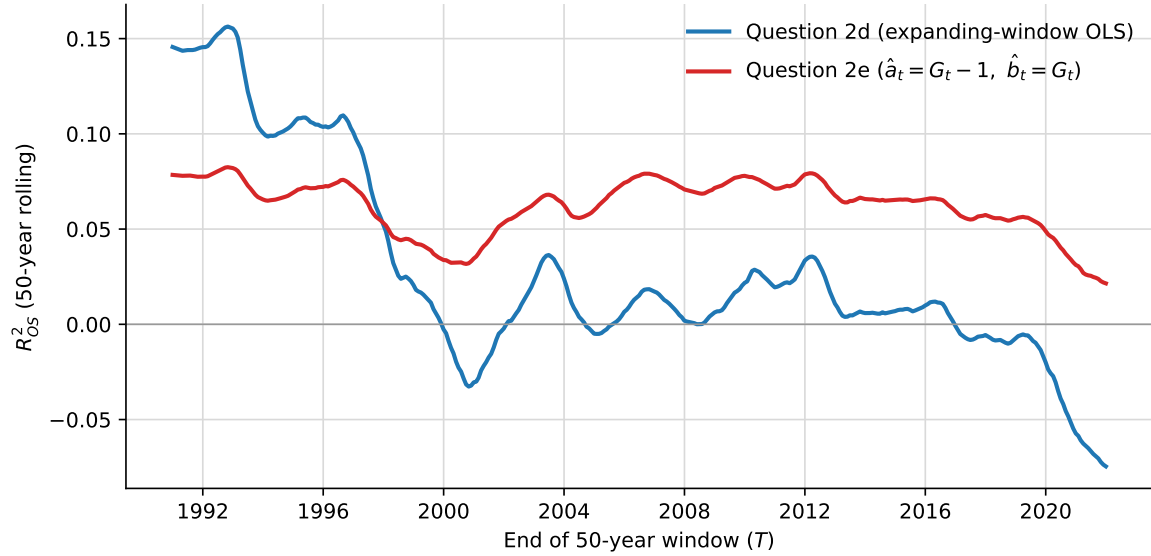


Figure 7: R_{OS}^2 over a rolling window of 600 forecast-target months ending at T , for T from 1990:12 to 2021:12, for the expanding-window OLS forecasts of Question 2d and the restricted forecasts of Question 2e ($\hat{a}_t = G_t - 1$, $\hat{b}_t = G_t$).

3 Question 3

3.1 Momentum signal construction and validation

Task. (1) Construct the momentum signal (MOM) monthly for the firms with data available in CRSP. Focus on the period from June of 1963 to the recent date to match the book-to-market construction of Questions (3b). As it is standard in the literature, restrict the analysis to common stocks of firms incorporated in the United States (CRSP `shrcd` = 10 or 11) trading on NYSE, Amex, or Nasdaq (CRSP `exchcd` = 1, 2 or 3), and exclude utilities ($4900 \leq SIC \leq 4949$) and financials ($6000 \leq SIC \leq 6999$).

The momentum signal of stock j at the end of month τ is

$$MOM_{j,\tau} = \prod_{m=\tau-11}^{\tau-1} (1 + R_{j,m}) - 1, \quad (37)$$

the cumulative return from the end of month $\tau - 12$ to the end of month $\tau - 1$, where $R_{j,m}$ is the CRSP monthly return (`MthRet`, which includes delisting returns). The returns are taken from every month of the CRSP file for the same PERMNO, whether or not the stock is in the universe in those months, and $MOM_{j,\tau}$ is missing when any of the 11 months has a missing return or no observation.

The signal months are 1963:06–2024:12. The universe at τ consists of common stocks of firms incorporated in the United States (CRSP fields `ShareType` = NS, `SecurityType` = EQTY, `SecuritySubType` = COM, `USIncFlg` = Y, `IssuerType` $\in$ {ACOR, CORP}, the equivalent of share codes 10 and 11) trading on NYSE, Amex or Nasdaq (`PrimaryExch` $\in$ {N, A, Q}, `ConditionalType` = RW, `TradingStatusFlg` = A), excluding utilities ($4900 \leq SIC \leq 4949$) and financials ($6000 \leq SIC \leq 6999$) and requiring non-missing market equity at τ . The SIC code is the CRSP `SICCD` at τ ; a code of zero is treated as missing and replaced by the most recent earlier non-zero code of the same PERMNO or, if there is none, by its next later one, and a stock that never has a non-zero code is kept.

Task. (2) To validate your MOM construction, merge your dataset with the CZ Dataset and define $MOM^{CZ} = \text{Mom12m}$ (use the column “permno” to identify the stock and the column “yyyymm” to identify the date). For each month in the sample, estimate a cross-firm regression of MOM^{CZ} onto your MOM and record the intercept and slope coefficients as well as the R^2 from these regressions. Provide three time series plots, one for the intercepts, one for the slopes, and one for the R^2 values.

For each month τ , the cross-firm regression

$$MOM_{j,\tau}^{CZ} = a_\tau + b_\tau MOM_{j,\tau} + \varepsilon_{j,\tau} \quad (38)$$

is estimated by OLS, where MOM^{CZ} is `Mom12m` from the CZ Dataset, merged on the stock identifier and the month τ , using the stocks in the universe at τ for which both signals are non-missing (2,428,767 stock-month observations in 739 months). Figures 8–10 plot $\hat{a}_\tau$, $\hat{b}_\tau$ and the R^2 of these regressions, and Table 6 summarizes them (code: `src/q3a.py`, `src/q3_common.py`).

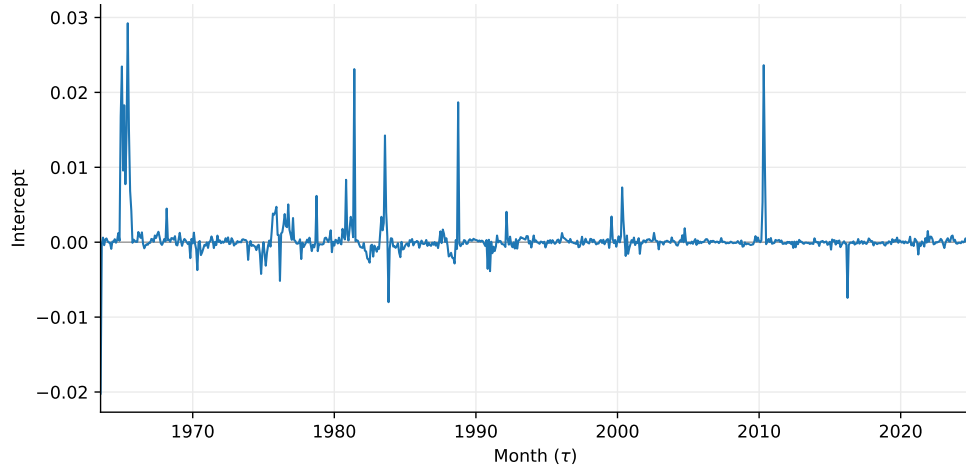


Figure 8: Intercept $\hat{a}_\tau$ of the monthly cross-firm regression (38), 1963:06–2024:12. The grey line marks 0.

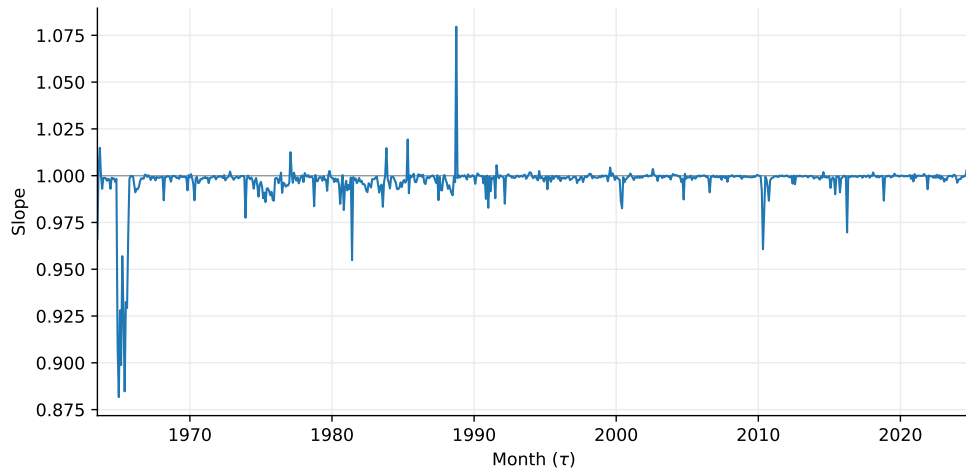


Figure 9: Slope $\hat{b}_\tau$ of the monthly cross-firm regression (38), 1963:06–2024:12. The grey line marks 1.

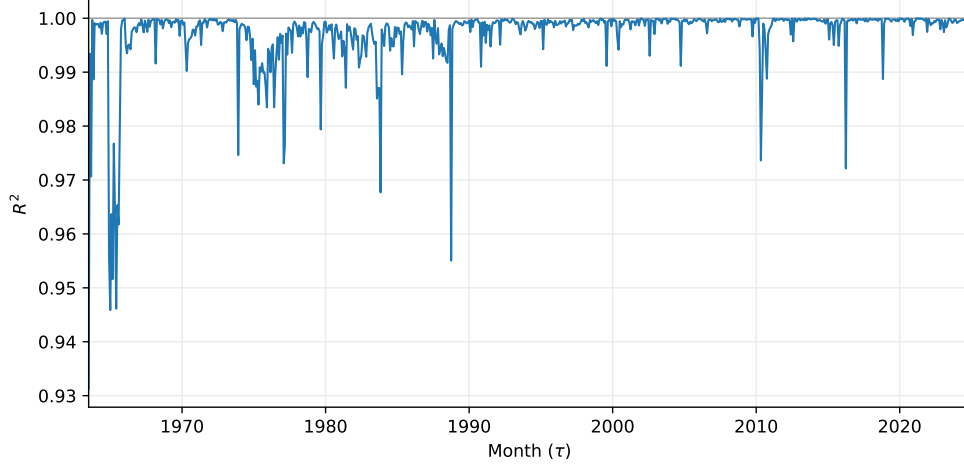


Figure 10: R^2 of the monthly cross-firm regression (38), 1963:06–2024:12. The grey line marks 1.

	Mean	Minimum	Maximum
Intercept	0.0003	-0.0202	0.0292
Slope	0.9972	0.8817	1.0796
R^2	0.9973	0.9313	1.0000
N	3,287	1,655	5,350

Table 6: Time-series mean, minimum and maximum of the intercept, slope, R^2 and number of stocks N of the monthly cross-firm regressions (38) of MOM^{CZ} on MOM , 739 signal months from 1963:06 to 2024:12.

3.2 Book-to-market signal construction and validation

Task. (1) Construct the book-to-market signal (BM) monthly for the firms with data available in the CRSP and COMPUSTAT datasets. To ensure all information used in the construction of BM is available to investors (see Fama and French (1992)), construct BM as of the end of June of year t using CRSP market equity data from December of calendar year $t - 1$ (for ME) and COMPUSTAT accounting data from the fiscal year ending in calendar year $t - 1$ (for BE). Then, keep BM fixed from June of year t to May of year $t + 1$, updating BM again at June of year $t + 1$. Also exclude firm-month observations with $BE \leq 0$. To avoid backfilling bias (see Fama and French (1993)), require a minimum of two previous years in COMPUSTAT for a company to be included in the analysis and construct BM from June of 1963 to the most recent date so that the first fiscal year of COMPUSTAT data is 1962. As it is standard in the literature, restrict the analysis to common stocks of firms incorporated in the United States (CRSP shrcd = 10 or 11) trading on NYSE, Amex, or Nasdaq (CRSP exchcd = 1, 2 or 3), and exclude utilities ($4900 \leq SIC \leq 4949$) and financials ($6000 \leq SIC \leq 6999$).

The book-to-market signal of stock j for year t is

$$BM_{j,t} = \frac{BE_{j,t-1}}{ME_{j,t-1}}, \quad (39)$$

where $BE_{j,t-1}$ is the book equity from the COMPUSTAT record whose fiscal year ends in calendar year $t - 1$ and $ME_{j,t-1} = |PRC| \times SHROUT$ is the CRSP market equity of the stock in December of year $t - 1$ (book equity is in millions and market equity in thousands of dollars, so the ratio is multiplied by 1000). $BM_{j,t}$ is assigned to the months τ from June of year t to May of year $t + 1$, for $t = 1963, \dots, 2024$, so the signal months are 1963:06–2024:12; the universe at τ is the one of Question 3a. $BM_{j,t}$ is missing when the stock has no CRSP observation or no market equity in December of $t - 1$.

Book equity is $BE = SE + TXDITC - BVPS$, where SE is SEQ, else CEQ + PSTK, else AT – LT; $TXDITC$ is TXDITC, or 0 when it is missing; and $BVPS$ is PSTKRV, else PSTKL, else PSTK, else 0. Observations with unavailable SE or with $BE \leq 0$ are excluded.

The accounting data come from the CRSP/COMPUSTAT Merged file, with link types LC and LU, primary links (LINKPRIM P or C), the fiscal year-end date inside the link window, and records reported in US dollars. When a stock has several records with a fiscal year-end in the same calendar year, a P link is preferred to a C link, and then the latest fiscal year-end date is used. To avoid backfilling bias, the record used must be preceded by at least two earlier records (distinct fiscal year-end dates) of the same firm in the file.

Task. (2) To validate your BE/ME construction, merge your dataset with the CZ Dataset and calculate $BM^{CZ} = \exp(\text{BMdec})$ (use the column “permno” to identify the stock and the column “yyyymm” to identify the date). For each month in the sample, estimate a cross-firm regression of BM^{CZ} onto your BM and record the intercept and slope coefficients as well as the R^2 from these regressions. Provide three time series plots, one for the intercepts, one for the slopes, and one for the R^2 values.

In the release of the CZ Dataset used here (2025.10), BMdec is the book-to-market ratio in levels and not its logarithm, so $BM^{CZ} = \text{BMdec}$ is used instead of $\exp(\text{BMdec})$. For each month τ , the cross-firm regression

$$BM_{j,\tau}^{CZ} = a_\tau + b_\tau BM_{j,\tau} + \varepsilon_{j,\tau} \quad (40)$$

is estimated by OLS, with BM^{CZ} merged on the stock identifier and the month τ , using the stocks in the universe at τ for which both signals are non-missing (1,803,835 stock-month observations in 739 months; between 454 and 3,914 stocks per month). Figures 11–13 plot $\hat{a}_\tau$, $\hat{b}_\tau$ and the R^2 of these regressions (code: `src/q3b.py`, `src/q3_common.py`).

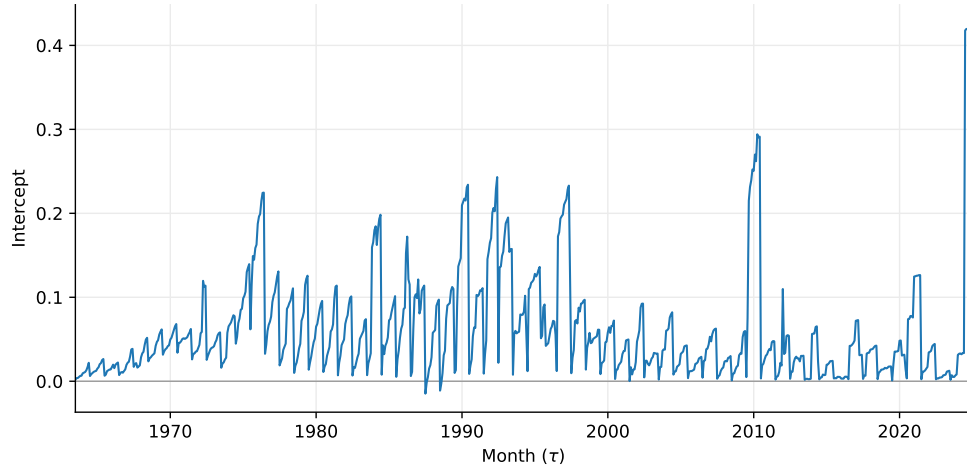


Figure 11: Intercept $\hat{a}_\tau$ of the monthly cross-firm regression (40), 1963:06–2024:12. The grey line marks 0.

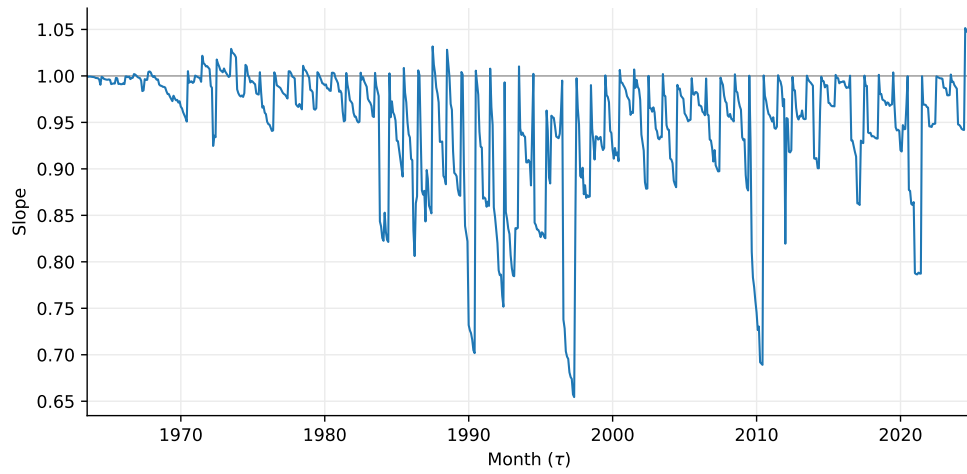


Figure 12: Slope $\hat{b}_\tau$ of the monthly cross-firm regression (40), 1963:06–2024:12. The grey line marks 1.

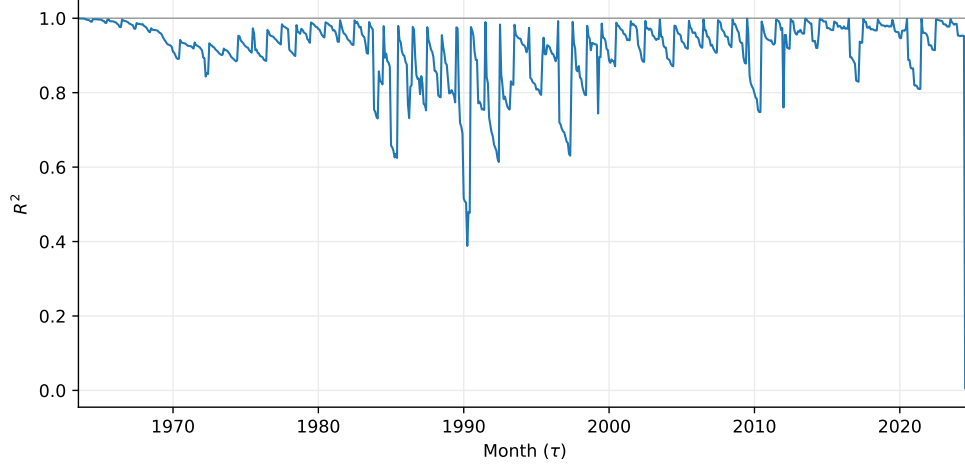


Figure 13: R^2 of the monthly cross-firm regression (40), 1963:06–2024:12. The grey line marks 1.

3.3 Decile portfolios

Task. (1) Let $GP^{CZ} = GP$. Then, for each of the three signals (BM^{CZ} , MOM^{CZ} , and GP^{CZ}), construct decile portfolios as described in the lectures (returns come from CRSP). Create five types of decile portfolios (for each signal):

- (i) Value-weighted, rebalanced annually (at June), and with NYSE breakpoints
- (ii) Equal-weighted, rebalanced annually (at June), and with NYSE breakpoints
- (iii) Value-weighted, rebalanced monthly, and with NYSE breakpoints
- (iv) Value-weighted, rebalanced annually (at June), and with general breakpoints
- (v) Equal-weighted, rebalanced monthly, and with general breakpoints

Decile portfolios are formed on each of the three signals of the CZ Dataset, $BM^{CZ} = \text{BMdec}$ (in levels, as in Question 3b), $MOM^{CZ} = \text{Mom12m}$ and $GP^{CZ} = GP$, dated τ . A stock enters a sort if it is in the universe at the formation date and its signal is non-missing and finite; for BM^{CZ} , stocks with $\text{BMdec} \leq 0$ are excluded.

With annual rebalancing, the portfolios are formed at the end of June of year t with the signal and the universe of June t and are held for the return months from July of year t to June of year $t + 1$; a stock without a signal in June t is left out until the next June. With monthly rebalancing, the portfolios are formed at the end of month τ with the signal and the universe of τ and are held for the return month $\tau + 1$. The breakpoints are the 10th, 20th, ..., 90th percentiles of the signal at the formation date, computed among the NYSE stocks in the sort (NYSE breakpoints) or among all stocks in the sort (general breakpoints); a stock is in decile d if its signal is above the $(d - 1)$ -th breakpoint and at or below the d -th, so that decile 1 has the lowest and decile 10 the highest signal.

The portfolio return of month $\tau + 1$ uses the members with a non-missing return in $\tau + 1$, with equal weights (EW) or with weights proportional to the market equity at the end of month τ (VW), which are updated every month, also within the annual holding period; a member with missing market equity at τ is excluded from the VW return of $\tau + 1$. Excess returns are $xR_{p,\tau+1} = R_{p,\tau+1} - R_{f,\tau+1}$, with the risk-free return from the Ken French data library. The five types of portfolios are: (i) VW, annual rebalancing, NYSE breakpoints; (ii) EW, annual rebalancing, NYSE breakpoints; (iii) VW, monthly rebalancing, NYSE breakpoints; (iv) VW, annual rebalancing, general breakpoints; (v) EW, monthly rebalancing, general breakpoints. The return months are

1963:07–2025:01 (739 months) for all five types; the last annual formation is June 2024 (code: `src/q3c.py`, `src/q3_common.py`).

Task. (2) For each of the five types of decile portfolios, provide a scatterplot with the decile value (1 to 10) in the x-axis and the average portfolio excess return in the y-axis. To keep the three signals in the same scatterplot (so that you have a total of five scatterplots), use the color blue for BM^{CZ} portfolios, red for MOM^{CZ} portfolios, and green for GP^{CZ} portfolios.

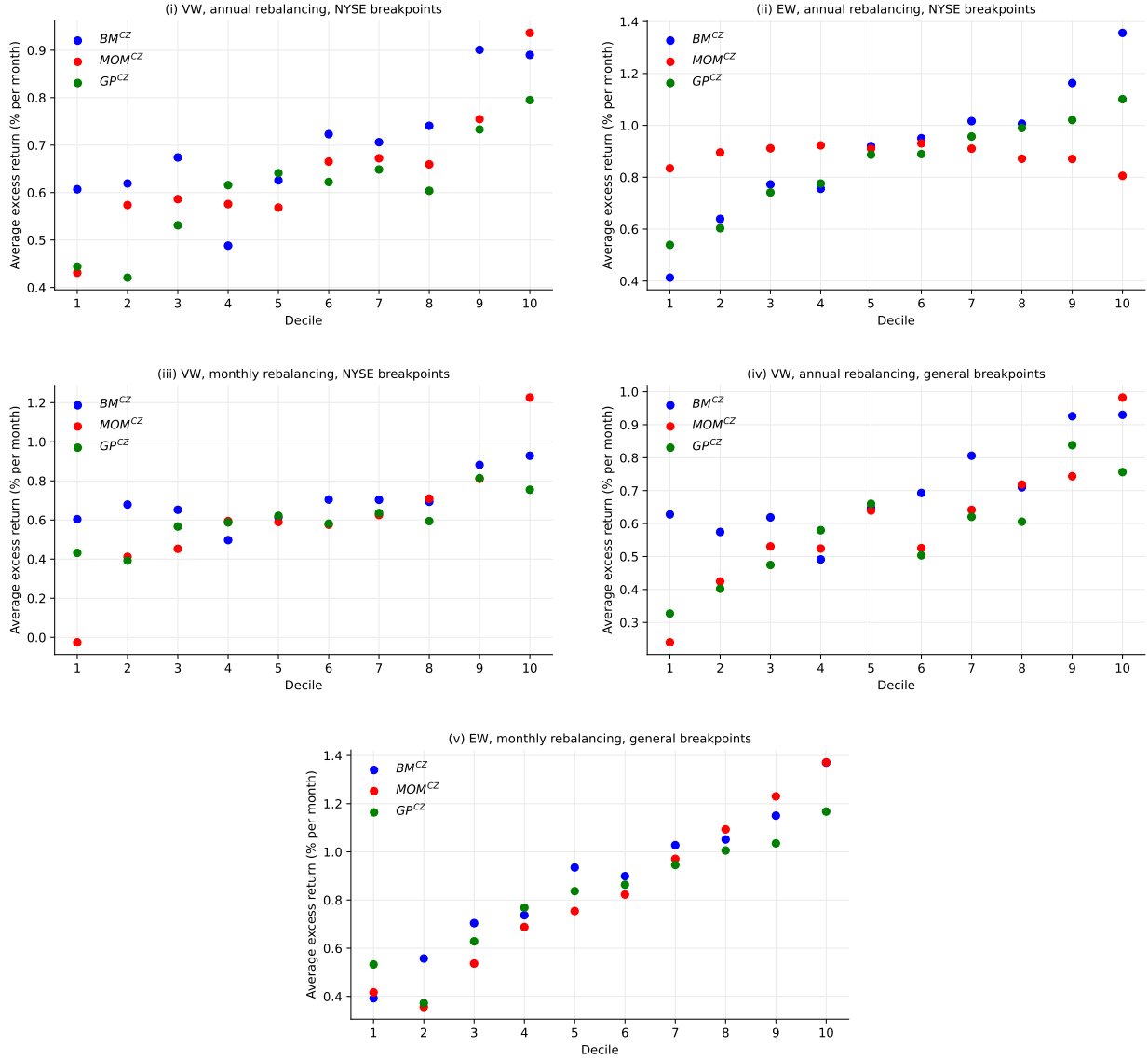


Figure 14: Average monthly excess return (in percent) of the decile portfolios against the decile, for the five types of portfolios; blue: BM^{CZ} , red: MOM^{CZ} , green: GP^{CZ} . Return months 1963:07–2025:01.

Task. (3) Moreover calculate the HML portfolio (decile 10 - decile 1) in each case (you should have a total of 15 HML portfolios) and report in a table the HML average returns as well as their respective t-statistics (using Newey and West (1987, 1994) standard errors, see Footnote 3).

Table 7 reports, for each type and signal, the average return of the HML portfolio, defined as decile 10 minus decile 1.

Type	Portfolios	BM^{CZ}	MOM^{CZ}	GP^{CZ}
(i)	VW, annual, NYSE breakpoints	0.283 [1.27]	0.505 [2.19]	0.351 [2.10]
(ii)	EW, annual, NYSE breakpoints	0.943 [4.18]	-0.029 [-0.14]	0.562 [3.29]
(iii)	VW, monthly, NYSE breakpoints	0.325 [1.51]	1.251 [4.62]	0.323 [1.89]
(iv)	VW, annual, general breakpoints	0.302 [1.13]	0.743 [2.58]	0.429 [2.05]
(v)	EW, monthly, general breakpoints	0.978 [4.12]	0.955 [3.15]	0.635 [2.98]

Table 7: Average returns of the HML portfolios (decile 10 minus decile 1), in percent per month, with t -statistics in brackets. The t -statistics come from a regression of the HML return on a constant with Newey–West (1987, 1994) standard errors, with the lag length selected by the automatic bandwidth rule as in method (v) of Question 2b (between 2 and 15 lags). Return months 1963:07–2025:01 (739 observations for each portfolio).

3.4 Fama–MacBeth regressions

Task. Estimate Fama–MacBeth regressions of $xR_{j,\tau+1}$ on signals available at month τ and report the results in a table (including the t -statistic of each coefficient). Estimate the following regression specifications (at each τ , use both OLS and WLS with month τ market-equity weights):

- (i) $xR_{j,\tau+1} = a + b_{BM} \cdot Q_{j,\tau}^{BM} + \varepsilon_{j,\tau+1}$
- (ii) $xR_{j,\tau+1} = a + b_{GP} \cdot Q_{j,\tau}^{GP} + \varepsilon_{j,\tau+1}$
- (iii) $xR_{j,\tau+1} = a + b_{Dur} \cdot Q_{j,\tau}^{Dur} + \varepsilon_{j,\tau+1}$
- (iv) $xR_{j,\tau+1} = a + b_{BM} \cdot Q_{j,\tau}^{BM} + b_{GP} \cdot Q_{j,\tau}^{GP} + \varepsilon_{j,\tau+1}$
- (v) $xR_{j,\tau+1} = a + b_{Dur} \cdot Q_{j,\tau}^{Dur} + b_{BM} \cdot Q_{j,\tau}^{BM} + \varepsilon_{j,\tau+1}$
- (vi) $xR_{j,\tau+1} = a + b_{Dur} \cdot Q_{j,\tau}^{Dur} + b_{GP} \cdot Q_{j,\tau}^{GP} + \varepsilon_{j,\tau+1}$
- (vii) $xR_{j,\tau+1} = a + b_{Dur} \cdot Q_{j,\tau}^{Dur} + b_{BM} \cdot Q_{j,\tau}^{BM} + b_{GP} \cdot Q_{j,\tau}^{GP} + \varepsilon_{j,\tau+1}$

where $Q_{j,\tau}^X$ reflects the quantile of signal $X_{j,\tau}$ in the cross-firm distribution of X at month τ (for BM and GP you should use the Chen and Zimmermann (2022) version of the signals).

The signals at month τ are $BM^{CZ} = \text{BMdec}$ and $GP^{CZ} = \text{GP}$ from the CZ Dataset, as in Question 3c, and the equity duration Dur from the Dur Dataset, where the value with $\text{FF.YEAR} = t$ is the signal for the months from June of year t to May of year $t + 1$, the timing of BM . A signal is valid if it is non-missing and finite; for BM^{CZ} , stocks with $\text{BMdec} \leq 0$ are excluded. The quantile of signal X is

$$Q_{j,\tau}^X = \frac{\text{rank}(X_{j,\tau})}{N_\tau^X}, \quad X \in \{BM^{CZ}, GP^{CZ}, Dur\}, \quad (41)$$

where the rank (1 for the lowest value, the average rank for ties) and the number of stocks N_τ^X are taken over all stocks in the universe at τ with a valid X , whether or not they enter a regression.

For each month τ , the following cross-firm regressions of the excess return $xR_{j,\tau+1} = R_{j,\tau+1} - R_{f,\tau+1}$, in percent per month, are estimated by OLS and by WLS with weights equal to the market equity $ME_{j,\tau}$:

$$\begin{aligned}
\text{(i)} \quad & xR_{j,\tau+1} = a + b_{BM} Q_{j,\tau}^{BM} + \varepsilon_{j,\tau+1}, \\
\text{(ii)} \quad & xR_{j,\tau+1} = a + b_{GP} Q_{j,\tau}^{GP} + \varepsilon_{j,\tau+1}, \\
\text{(iii)} \quad & xR_{j,\tau+1} = a + b_{Dur} Q_{j,\tau}^{Dur} + \varepsilon_{j,\tau+1}, \\
\text{(iv)} \quad & xR_{j,\tau+1} = a + b_{BM} Q_{j,\tau}^{BM} + b_{GP} Q_{j,\tau}^{GP} + \varepsilon_{j,\tau+1}, \\
\text{(v)} \quad & xR_{j,\tau+1} = a + b_{Dur} Q_{j,\tau}^{Dur} + b_{BM} Q_{j,\tau}^{BM} + \varepsilon_{j,\tau+1}, \\
\text{(vi)} \quad & xR_{j,\tau+1} = a + b_{Dur} Q_{j,\tau}^{Dur} + b_{GP} Q_{j,\tau}^{GP} + \varepsilon_{j,\tau+1}, \\
\text{(vii)} \quad & xR_{j,\tau+1} = a + b_{Dur} Q_{j,\tau}^{Dur} + b_{GP} Q_{j,\tau}^{GP} + b_{BM} Q_{j,\tau}^{BM} + \varepsilon_{j,\tau+1}.
\end{aligned} \tag{42}$$

Each regression uses the stocks in the universe at τ for which all regressors of the specification are valid and $xR_{j,\tau+1}$ is non-missing. The signal months are 1963:06–2024:12 for specifications (i), (ii) and (iv), and 1973:06–2024:12 for specifications (iii), (v), (vi) and (vii), because *Dur* starts in 1973. The Fama–MacBeth estimate is the time-series mean of the monthly coefficient estimates, and its t -statistic comes from a regression of the monthly coefficient series on a constant with Newey–West (1987, 1994) standard errors, with the lag length selected by the automatic bandwidth rule as in method (v) of Question 2b. Since xR is in percent per month, a coefficient is the difference in percent per month between the highest and the lowest quantile (code: `src/q3d.py`, `src/q3_common.py`).

		a	b_{BM}	b_{GP}	b_{Dur}	Avg. N	Months
(i)	OLS	0.410 [1.55]	0.944 [4.20]			2,695	739
	WLS	0.572 [2.85]	0.193 [0.75]			2,695	739
(ii)	OLS	0.442 [1.47]		0.753 [4.14]		3,015	739
	WLS	0.330 [1.60]		0.494 [2.51]		3,015	739
(iii)	OLS	1.610 [7.18]			-1.200 [-6.03]	2,364	619
	WLS	1.337 [7.17]			-0.912 [-3.01]	2,364	619
(iv)	OLS	-0.093 [-0.30]	1.074 [4.75]	0.849 [4.89]		2,626	739
	WLS	0.009 [0.03]	0.634 [2.46]	0.752 [4.11]		2,626	739
(v)	OLS	1.407 [5.93]	0.246 [0.88]		-1.055 [-4.06]	2,339	619
	WLS	1.304 [3.60]	-0.176 [-0.49]		-0.815 [-2.28]	2,339	619
(vi)	OLS	1.438 [4.99]		0.243 [1.35]	-1.129 [-5.18]	2,310	619
	WLS	0.789 [2.04]		0.564 [1.74]	-0.551 [-1.32]	2,310	619
(vii)	OLS	0.930 [3.42]	0.470 [1.43]	0.422 [1.98]	-0.795 [-2.68]	2,302	619
	WLS	0.450 [1.19]	0.306 [1.12]	0.637 [2.32]	-0.260 [-0.68]	2,302	619

Table 8: Fama–MacBeth estimates of regressions (42), in percent per month, with Newey–West (1987, 1994) t -statistics in brackets (between 2 and 16 lags). OLS and WLS refer to the monthly cross-firm regressions; the WLS weights are the market equity at τ . Avg. N is the average number of stocks in the monthly regressions.

3.5 Portfolio-level multivariate regressions

Task. Gonçalves (2021b) proposes the use of multivariate regressions for portfolios analogous to the use of firm-level multivariate regressions in Question (3d). For instance, if we have 10 BM^{CZ} decile portfolios and 10 GP^{CZ} decile portfolios, then for each portfolio p we can calculate the average BM^{CZ} decile of its firms (let's call it $Dec_{p,\tau}^{BM}$) as well as the average GP^{CZ} decile of its firms (let's call it $Dec_{p,\tau}^{GP}$) and estimate the panel regression $xR_{p,\tau+1} = a + b_{BM} \cdot Dec_{p,\tau}^{BM} + b_{GP} \cdot Dec_{p,\tau}^{GP} + \varepsilon_{p,\tau+1}$ by pooled OLS using the 20 portfolios with Driscoll and Kraay (1998) standard errors. Following this approach, estimate the following regression specifications and report the results in a table (including t-statistics):

- (i) $xR_{p,\tau+1} = a + b_{BM} \cdot Dec_{p,\tau}^{BM} + \varepsilon_{p,\tau+1}$
- (ii) $xR_{p,\tau+1} = a + b_{GP} \cdot Dec_{p,\tau}^{GP} + \varepsilon_{p,\tau+1}$
- (iii) $xR_{p,\tau+1} = a + b_{Dur} \cdot Dec_{p,\tau}^{Dur} + \varepsilon_{p,\tau+1}$
- (iv) $xR_{p,\tau+1} = a + b_{BM} \cdot Dec_{p,\tau}^{BM} + b_{GP} \cdot Dec_{p,\tau}^{GP} + \varepsilon_{p,\tau+1}$
- (v) $xR_{p,\tau+1} = a + b_{Dur} \cdot Dec_{p,\tau}^{Dur} + b_{BM} \cdot Dec_{p,\tau}^{BM} + \varepsilon_{p,\tau+1}$
- (vi) $xR_{p,\tau+1} = a + b_{Dur} \cdot Dec_{p,\tau}^{Dur} + b_{GP} \cdot Dec_{p,\tau}^{GP} + \varepsilon_{p,\tau+1}$
- (vii) $xR_{p,\tau+1} = a + b_{Dur} \cdot Dec_{p,\tau}^{Dur} + b_{BM} \cdot Dec_{p,\tau}^{BM} + b_{GP} \cdot Dec_{p,\tau}^{GP} + \varepsilon_{p,\tau+1}$

Estimate each specification above using value-weighted portfolios as well as equal-weighted portfolios (in both cases the portfolios are rebalanced annually and rely on NYSE breakpoints). Moreover, for the BM and GP portfolios use the Chen and Zimmermann (2022) version of the signals.

The portfolios are decile portfolios sorted on BM^{CZ} , on GP^{CZ} and on Dur , rebalanced annually with NYSE breakpoints, value-weighted or equal-weighted. For BM^{CZ} and GP^{CZ} they are the portfolios of types (i) and (ii) of Question 3c; the Dur portfolios are built with the same rules, sorted at the end of June of year t on the duration with $FF.YEAR = t$. The signals and their validity are those of Questions 3c and 3d.

For each signal X , the decile of a stock in X is its decile in the annual NYSE sort on X at the end of June of year t , fixed for the return months from July of year t to June of year $t + 1$. For portfolio p and return month $\tau + 1$, the regressor $Dec_{p,\tau}^X$ is the average of the deciles in X over the members used for the return of $\tau + 1$ that have a decile in X , with the weights of the portfolio return (market equity at the end of τ for the value-weighted portfolios, equal weights for the equal-weighted portfolios) renormalized over those members. For the signal on which the portfolio is sorted, $Dec_{p,\tau}^X$ is the decile of the portfolio.

The following regressions of the portfolio excess return $xR_{p,\tau+1}$, in percent per month, are estimated by pooled OLS:

$$\begin{aligned}
 \text{(i)} \quad & xR_{p,\tau+1} = a + b_{BM} Dec_{p,\tau}^{BM} + \varepsilon_{p,\tau+1}, \\
 \text{(ii)} \quad & xR_{p,\tau+1} = a + b_{GP} Dec_{p,\tau}^{GP} + \varepsilon_{p,\tau+1}, \\
 \text{(iii)} \quad & xR_{p,\tau+1} = a + b_{Dur} Dec_{p,\tau}^{Dur} + \varepsilon_{p,\tau+1}, \\
 \text{(iv)} \quad & xR_{p,\tau+1} = a + b_{BM} Dec_{p,\tau}^{BM} + b_{GP} Dec_{p,\tau}^{GP} + \varepsilon_{p,\tau+1}, \\
 \text{(v)} \quad & xR_{p,\tau+1} = a + b_{Dur} Dec_{p,\tau}^{Dur} + b_{BM} Dec_{p,\tau}^{BM} + \varepsilon_{p,\tau+1}, \\
 \text{(vi)} \quad & xR_{p,\tau+1} = a + b_{Dur} Dec_{p,\tau}^{Dur} + b_{GP} Dec_{p,\tau}^{GP} + \varepsilon_{p,\tau+1}, \\
 \text{(vii)} \quad & xR_{p,\tau+1} = a + b_{Dur} Dec_{p,\tau}^{Dur} + b_{GP} Dec_{p,\tau}^{GP} + b_{BM} Dec_{p,\tau}^{BM} + \varepsilon_{p,\tau+1}.
 \end{aligned} \tag{43}$$

Each specification stacks the portfolios sorted on the signals that appear in it: 10 portfolios for (i)–(iii), 20 for (iv)–(vi) and 30 for (vii). The return months are 1963:07–2025:01 for specifications

(i), (ii) and (iv), and 1973:07–2025:01 for specifications (iii), (v), (vi) and (vii), because Dur starts in 1973. The standard errors are those of Driscoll and Kraay (1998) with the Bartlett kernel, with the lag length selected by the Newey–West (1994) automatic bandwidth rule of method (v) of Question 2b, applied to the moments summed across portfolios within each month. Since xR is in percent per month, a coefficient is in percent per month per decile (code: `src/q3e.py`, `src/q3c.py`, `src/q3_common.py`).

	a	b_{BM}	b_{GP}	b_{Dur}	Portfolios	Months
Value-weighted portfolios						
(i)	0.511 [2.71]	0.034 [1.61]			10	739
(ii)	0.413 [2.00]		0.035 [2.25]		10	739
(iii)	1.184 [6.39]			-0.064 [-3.12]	10	619
(iv)	-0.224 [-0.61]	0.088 [2.48]	0.089 [2.84]		20	739
(v)	1.280 [5.36]	-0.003 [-0.09]		-0.082 [-3.30]	20	619
(vi)	1.383 [4.69]		-0.008 [-0.34]	-0.098 [-3.38]	20	619
(vii)	1.731 [2.41]	-0.027 [-0.49]	-0.024 [-0.53]	-0.121 [-2.85]	30	619
Equal-weighted portfolios						
(i)	0.428 [1.69]	0.086 [4.52]			10	739
(ii)	0.525 [1.84]		0.059 [4.44]		10	739
(iii)	1.542 [6.91]			-0.099 [-6.03]	10	619
(iv)	-0.420 [-1.24]	0.128 [5.50]	0.106 [6.04]		20	739
(v)	1.308 [4.65]	0.035 [0.98]		-0.094 [-3.11]	20	619
(vi)	1.574 [5.29]		0.005 [0.26]	-0.117 [-5.84]	20	619
(vii)	0.879 [1.40]	0.055 [1.13]	0.043 [1.23]	-0.083 [-1.97]	30	619

Table 9: Pooled OLS estimates of regressions (43) for value-weighted and equal-weighted decile portfolios (annual rebalancing, NYSE breakpoints), in percent per month per decile, with Driscoll–Kraay (1998) t -statistics in brackets (Bartlett kernel, between 1 and 11 lags).

4 Question 4

4.1 Yields, forward rates, and returns

Task. The “Bond Dataset” contains yields in percentual terms, $Y_{b,t}^{(H)} \cdot 100\%$. Construct time series of:

- (i) Log yields: $y_{b,t}^{(H)} = \log(1 + Y_{b,t}^{(H)})$
 - (ii) Log forward rates: $f_{b,t}^{(H)} = H \cdot y_{b,t}^{(H)} - (H - 1) \cdot y_{b,t}^{(H-1)}$
 - (iii) Log annual returns: $r_{b,t}^{(H)} = H \cdot y_{b,t-1}^{(H)} - (H - 1) \cdot y_{b,t}^{(H-1)}$
- For $H = 2, 3, 4, 5$, calculate and report in a table the average values of

- (i) $xy_{b,t}^{(H)} = y_{b,t}^{(H)} - y_{b,t}^{(1)}$
- (ii) $xf_{b,t}^{(H)} = f_{b,t}^{(H)} - f_{b,t}^{(1)}$
- (iii) $xr_{b,t}^{(H)} = r_{b,t}^{(H)} - r_{b,t}^{(1)}$

From the Fama–Bliss yields $Y_{b,t}^{(H)}$ (in percent) for $H = 1, \dots, 5$, the log yields, log forward rates and log annual returns are

$$y_{b,t}^{(H)} = \log\left(1 + Y_{b,t}^{(H)}/100\right), \quad f_{b,t}^{(H)} = H y_{b,t}^{(H)} - (H-1) y_{b,t}^{(H-1)}, \quad r_{b,t}^{(H)} = H y_{b,t-1}^{(H)} - (H-1) y_{b,t}^{(H-1)}, \quad (44)$$

with $y^{(0)} = 0$, so that $f_{b,t}^{(1)} = y_{b,t}^{(1)}$ and $r_{b,t}^{(1)} = y_{b,t-1}^{(1)}$; $t - 1$ is 12 months before t . The spreads are $xy^{(H)} = y^{(H)} - y^{(1)}$, $xf^{(H)} = f^{(H)} - f^{(1)}$ and $xr^{(H)} = r^{(H)} - r^{(1)}$ (code: `src/q4a.py`).

Average (%)	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$xy_{b,t}^{(H)}$ (yield spread)	0.169	0.331	0.471	0.568
$xf_{b,t}^{(H)}$ (forward spread)	0.339	0.654	0.892	0.956
$xr_{b,t}^{(H)}$ (excess return)	0.315	0.607	0.820	0.872
Common sample 1953:06–2024:12				

Table 10: Average yield spreads, forward spreads and excess log returns of H -year Fama–Bliss discount bonds over the 1-year bond, $H = 2, \dots, 5$, in percent per year. Monthly observations over the common sample on which all three series are defined, 1953:06–2024:12.

4.2 Hold-to-maturity excess return regressions

Task. For $H = 2, 3, 4, 5$, estimate the regression $(1/H) \cdot xr_{b,t \rightarrow t+H}^{(H)} = a^{(H)} + b^{(H)} \cdot xy_{b,t}^{(H)} + \varepsilon_t^{(H)}$ (4.1) where the hold-to-maturity excess return is $xr_{b,t \rightarrow t+H}^{(H)} = \sum_{h=1}^H xr_{t+h}^{(H-h+1)} = \sum_{h=1}^H (r_{t+h}^{(H-h+1)} - r_{t+h}^{(1)})$. Using the results from the regressions in Equation 4.1, reproduce a table analogous to the first panel of the first table on slide 5.5 of the Module 1 lecture notes. For the t-statistics, use the method in Hansen and Hodrick (1980) (see Footnote 3).

Table 11 reports OLS estimates of

$$\frac{1}{H} xr_{b,t \rightarrow t+H}^{(H)} = a^{(H)} + b^{(H)} xy_{b,t}^{(H)} + \varepsilon_t^{(H)}, \quad H = 2, \dots, 5, \quad (45)$$

where the hold-to-maturity excess return is

$$xr_{b,t \rightarrow t+H}^{(H)} = \sum_{h=1}^H xr_{b,t+h}^{(H-h+1)} = \sum_{h=1}^H \left(r_{b,t+h}^{(H-h+1)} - r_{b,t+h}^{(1)} \right), \quad (46)$$

with r and xr as in (44), $t+h$ being 12h months after t , and the $h=H$ term identically zero because $xr_{b,t}^{(1)} = 0$. Each regression uses every month t for which all non-zero terms of (46) are observed, i.e. $t + 12(H-1) \leq 2024:12$. The t -statistics use Hansen–Hodrick (1980) standard errors with $L = 12H - 1$ lags (code: `src/q4b.py`).

	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$b^{(H)}$	0.677	0.532	0.414	0.345
	[3.56]	[3.14]	[2.46]	[2.25]
R^2 (%)	7.7	5.2	3.8	3.2

Table 11: OLS estimates of $b^{(H)}$ in regression (45), with Hansen–Hodrick (1980) t -statistics ($12H-1$ lags) in brackets and R^2 in percent. Monthly observations from 1952:06; the last predictor month is 2023:12, 2022:12, 2021:12 and 2020:12 for $H = 2, 3, 4, 5$ (859, 847, 835 and 823 observations).

4.3 One-year excess return regressions on forward spreads

Task. For $H = 2, 3, 4, 5$, estimate the regression $xr_{b,t+1}^{(H)} = a^{(H)} + b^{(H)} \cdot xf_{b,t}^{(H)} + \varepsilon_t^{(H)}$ (4.2). Using the results from the regressions in Equation 4.2, reproduce a table analogous to the second panel of the first table on slide 5.5 of the Module 1 lecture notes. For the t -statistics, use the method in Newey and West (1987, 1994) (see Footnote 3).

Table 12 reports OLS estimates of

$$xr_{b,t+1}^{(H)} = a^{(H)} + b^{(H)} xf_{b,t}^{(H)} + \varepsilon_t^{(H)}, \quad H = 2, \dots, 5, \quad (47)$$

with xr and xf as in (44) and $t+1$ being 12 months after t . Each regression uses every month t for which $xr_{b,t+1}^{(H)}$ is observed, i.e. $t + 12 \leq 2024:12$, so all four samples are 1952:06–2023:12. The t -statistics use Newey–West (1987) standard errors with the lag length selected by the Newey–West (1994) automatic bandwidth rule, as in method (v) of Question 2b (code: `src/q4c.py`).

	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$b^{(H)}$	0.677	0.889	1.116	0.964
	[3.23]	[3.35]	[3.63]	[2.92]
R^2 (%)	7.7	8.6	10.7	6.7

Table 12: OLS estimates of $b^{(H)}$ in regression (47), with Newey–West (1987, 1994) t -statistics in brackets (selected lags 23, 23, 24 and 23 for $H = 2, 3, 4, 5$) and R^2 in percent. Monthly observations, predictor months 1952:06–2023:12 (859 observations for each H).

4.4 Cochrane–Piazzesi factor

Task. Estimate the Cochrane and Piazzesi (2005) factor from

$$\frac{1}{4} \cdot \sum_{H=2}^5 xr_{b,t+1}^{(H)} = \theta_0 + \underbrace{\sum_{H=1}^5 \theta_H \cdot f_{b,t}^{(H)}}_{cp_t} + u_t \quad (4.3)$$

and provide a time-series plot of cp_t with NBER recession gray areas in the background (the NBER recession dummies are available under <https://fred.stlouisfed.org/series/USREC>)

The regression

$$\frac{1}{4} \sum_{H=2}^5 xr_{b,t+1}^{(H)} = \theta_0 + \sum_{H=1}^5 \theta_H f_{b,t}^{(H)} + u_t \quad (48)$$

is estimated by OLS, with xr and f as in (44) and $t + 1$ being 12 months after t . It uses every month t for which the dependent variable is observed, i.e. $t + 12 \leq 2024:12$, so the sample is 1952:06–2023:12 (859 observations; $R^2 = 0.158$). The Cochrane–Piazzesi factor is the fitted value $cp_t = \hat{\theta}_0 + \sum_{H=1}^5 \hat{\theta}_H f_{b,t}^{(H)}$, defined on the regression months only. Figure 15 plots cp_t , with the NBER recession months (FRED series USREC, 1 for a recession month) shaded in gray (code: `src/q4d.py`).

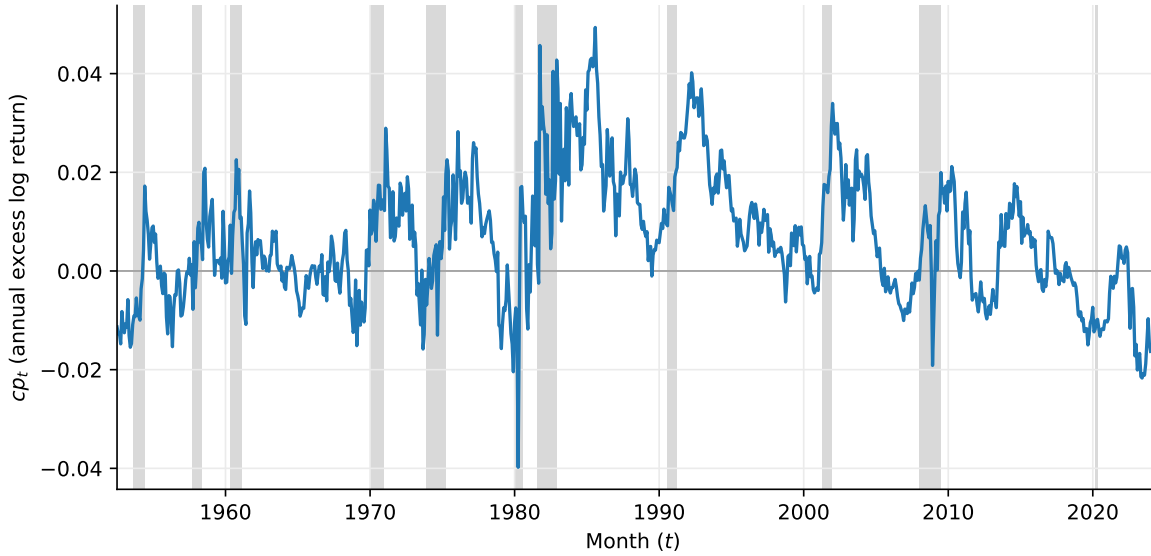


Figure 15: Cochrane–Piazzesi factor cp_t , the fitted value of regression (48), 1952:06–2023:12. Gray areas are NBER recessions.

4.5 Excess return regressions on the CP factor

Task. For $H = 2, 3, 4, 5$, estimate the regression $xr_{b,t+1}^{(H)} = a^{(H)} + b^{(H)} \cdot cp_t + \varepsilon_t^{(H)}$ (4.4). Using the results from the regressions in Equation 4.4, reproduce a table analogous to the second table on slide 5.5 of the Module 1 lecture notes. For the t -statistics, use the method in Newey and West (1987, 1994) (see Footnote 3).

Table 13 reports OLS estimates of

$$xr_{b,t+1}^{(H)} = a^{(H)} + b^{(H)} cp_t + \varepsilon_t^{(H)}, \quad H = 2, \dots, 5, \quad (49)$$

with xr as in (44), cp_t from Question 4d and $t+1$ being 12 months after t . Each regression uses every month t for which $xr_{b,t+1}^{(H)}$ is observed, i.e. $t+12 \leq 2024:12$, so all four samples are 1952:06–2023:12, the months on which cp_t is defined. The t -statistics use Newey–West (1987) standard errors with the lag length selected by the Newey–West (1994) automatic bandwidth rule, as in method (v) of Question 2b (code: `src/q4e.py`).

	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$b^{(H)}$	0.442	0.827	1.252	1.479
	[4.17]	[4.19]	[4.45]	[4.26]
R^2 (%)	13.7	14.4	17.0	15.5

Table 13: OLS estimates of $b^{(H)}$ in regression (49), with Newey–West (1987, 1994) t -statistics in brackets (23 selected lags for every H) and R^2 in percent. Monthly observations, predictor months 1952:06–2023:12 (859 observations for each H).